METRIC
MIL-L-24732(NAVY)
31 August 1989

MILITARY SPECIFICATION

LIGHT SOURCE, RIGID AND FLEXIBLE FIBERSCOPE, FIBER OPTIC (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for light sources (see 6.5.1) for rigid and flexible fiberscopes. The light source is used to produce effective illumination over the fiberscopes' field of view during nondestructive inspections of equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6032
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

J-C-580	- Cord, Electrical and Wire, Electrical (0- to 600-Volt Service).
TT-I-735	- Isopropyl Alcohol.
PPP-C-843	- Cushioning Material, Cellulosic.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-C-4150	- Cases, Transit and Storage, Waterproof and Water-Vaporproof.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-H-17672	- Hydraulic Fluid, Petroleum, Inhibited.
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-L-24732/1	- Light Source, Fiberscope, Fiber Optic, 150 Watt (Metric).

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-313	- Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.
FED-STD-595	- Colors.

MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-471	- Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-883	- Test Methods and Procedures for Microelectronics.
MIL-STD-889	- Dissimilar Metals.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:33Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Reliability (see 4.6.2). The light source shall be designed to have a lower test Mean-Time-Between-Failures (MTBF O_1) of 500 hours and an upper Mean-Time-Between-Failures (MTBF O_o) of 1,110 hours. The designed service lifetime shall be 10 years. During a 4-hour duty cycle, the light intensity shall be not less than 95 percent of the light source's maximum rated intensity.

3.4 Materials. The units shall be constructed of materials that will not produce toxic, corrosive, or explosive by-products. Materials shall not have adverse effects upon operational or maintenance personnel under all operational and environmental conditions, nor cause degradation of equipment performance. Materials shall be non-magnetic.

3.4.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

3.4.2 Metals. Metals, other than current carrying parts, shall be corrosion-resistant. All exposed corrosion-resistant steel parts shall have a passivated surface that is compatible with external coatings or platings. Cadmium plating shall not be used.

3.4.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably finished to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-889.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

3.4.4 Nonmetallic materials. Nonmetallic materials used in the construction of light sources shall not be degraded under the specified environmental conditions.

3.4.5 Material Safety Data Sheet (see 6.6). The contracting activity shall be provided a Material Safety Data Sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

3.5 Design and construction. The configuration and physical dimensions of the light source shall be as specified herein. The light source assembly shall be comprised of the light source enclosure, with carrying handle and shoulder strap, fiber optics light guide cable socket (female), power on/off switch, two power cables, dual light bulbs, lamp selector switch, power-on indication light, brightness or intensity control knob, cooling fan, overload fuse and voltage control switch or selector. The light source shall be compatible with all rigid and flexible fiberscope light guide cable connectors or as specified (see 3.1). Light sources with the same part number shall be physically and functionally interchangeable.

3.5.1 Power switch (see 4.6.1.1). The light source shall have a positive on/off switch.

3.5.2 Power cables (see 4.6.1.1). The power cables shall be not less than 3 meters long and able to withstand repeated coiling. A standard three-prong plug having two parallel blades shall be securely attached to the power cable. The ground lead and power cords shall securely attach and shall have a positive connection to the light source assembly in accordance with J-C-580. Two adapters shall be provided for military and European use as specified. The ground lead shall be securely grounded when the adapters are connected.

3.5.3 Lamp life/dual lamp capacity with lamp selection switch (see 4.6.1.1). Light sources shall have dual bulb capacity to provide near uninterrupted operation through manual switching. Direct access to change or switch bulbs without disassembling the unit, such as a rotating bulb system, shall be provided. A spare lamp, stored in the light source, shall be provided. Lamp life shall be as specified (see 3.1).

3.5.4 Light intensity control (see 4.6.1.1). A mechanical rheostat or shutter action control shall be used to change light intensity (see 6.5.2). A light intensity (brightness) control with output adjustment indicator permanently attached and marked to show zero, one-quarter, one-half, three-quarters, and full intensity, shall be provided.

3.5.5 Fusing (see 4.6.1.1). The light source assembly shall have a circuit fuse (10-ampere maximum) to protect against overload, shock and provide system safety. One fuse shall be provided as a spare and stored in the light source case.

3.5.6 Cooling fan (see 4.6.1.1). Forced-air cooling system shall be provided to regulate temperature and protect against lamp compartment overheating. The fan exhaust shall be vented to the atmosphere.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:33Z
Check the source to verify that this is the current version before use.

3.5.7 Light guide cable socket (receptacle) (see 4.6.1.1). The light guide cable socket shall ensure positive connection with the light guide cable connector (plug). A force of 26.5 ± 2.2 newtons shall be required to mate, and a force of 43.6 ± 2.2 newtons shall be required to demate the light source from the light guide. The dimensions of the socket shall be in accordance with figure 1.

3.5.8 Portable battery pack (see 4.6.1.1). A rechargeable, portable 12-volt battery pack kit shall be provided. The battery pack shall provide power for 1.5 hours of continuous operation. Included in the kit shall be a battery charger and two sets of batteries.

3.5.9 Carrying case (see 4.6.1.1). The carrying case shall be in accordance with MIL-C-4150.

3.5.10 Welding and brazing (see 4.6.1.1). Welding and brazing shall be in accordance with MIL-STD-454, requirements 24 and 59.

3.5.11 Terminals.

3.5.11.1 Terminal strength (see 4.6.3.1). Terminals shall reveal no evidence of breakage, loosening, or relative motion between the electrical terminals and the body of the light source.

3.5.11.2 Solderability (see 4.6.3.2). Terminal solderability shall be in accordance with MIL-STD-454, requirement 5.

3.5.12 Threads (see 4.6.1.1). Threads shall be in accordance with FED-STD-H28 or as specified (see 3.1).

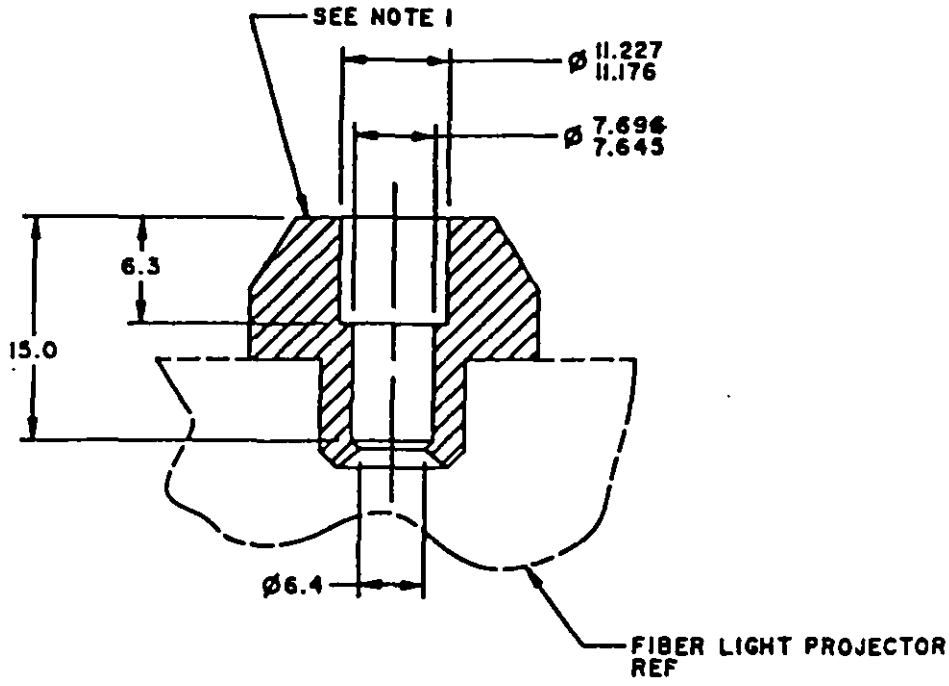
3.5.13 Weight (see 4.6.3.3). The weight of the light source and battery pack, together, shall be as specified (see 3.1).

3.5.14 Dimensions (see 4.6.1.1). The dimensions of the light source shall be as specified (see 3.1).

3.5.15 Color (see 4.6.1.1). The exterior of the light source carrying case shall be an orange-yellow color paint and have a nongloss, smooth finish in accordance with FED-STD-595.

3.5.16 Maintainability (see 4.6.3.4). The light source shall be designed to have a Mean-Time-to-Repair (MTTR) of 1-hour.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.



DIMENSIONS ARE IN MILLIMETERS

ALL MACHINED HOLES:

THRU 3.18 mm	+0.13 mm -0.05 mm
OVER 3.18 mm THRU 38.10 mm	+0.30 mm -0.08 mm
OVER 38.10 mm	+0.51 mm -0.10 mm

ALL SURFACES 3175/ µ mm

FIGURE 1. Light guide cable socket (receptacle).

Source: https://assist.dla.mil -- Downloaded: 20261O-20T01:33Z
Check the source to verify that this is the current version before use.

3.6 Performance characteristics.

3.6.1 Power requirements (see 4.6.4.1). The light source shall operate from a single phase 115-volt (plus or minus 10 percent), 60 hertz (Hz) (plus or minus 5 percent), U.S. commercial power alternating current (ac), or single phase 110-volt (plus or minus 8 percent), 400 Hz (plus or minus 5 percent) military ac power. Two hundred twenty volts (plus or minus 10 percent), 50 Hz (plus or minus 5 percent) European power supply and 12 volt/28 volt direct current (dc) battery power are options if specified (see 3.1). The light source shall meet all performance requirements specified herein while operating with specified power supply voltages and their tolerances. The total light source power consumption shall be as specified (see 3.1).

3.6.2 Light intensity. The light source shall be capable of varying the light intensity at the fiberscope distal end. The light source shall furnish cold white light illumination and shall operate continuously at maximum intensity for a minimum of 4 hours. Unless otherwise specified (see 3.1), the light source shall be a bulb which requires not less than 150 watts.

3.7 Environmental requirements. The light source shall meet all requirements specified under the following environmental conditions.

3.7.1 Environmental temperature extremes (see 4.6.5.1). The light source shall be fully functional and show no evidence of physical degradation. The light source environmental temperature range shall be as specified (see 3.1), in accordance with the ranges in table I.

TABLE I. Environmental temperature ranges.

Range	Operating (°C)	Storage (°C)
1	-54 to +85	-62 to +85
2	-28 to +85	-62 to +85

3.7.2 Thermal shock (see 4.6.5.2). The light source shall be fully functional and show no evidence of physical degradation. The temperature extremes shall be the specified storage temperature extremes (see 3.7.1).

3.7.3 Humidity (see 4.6.5.3). The light source shall show no evidence of deterioration of parts or materials, loosening of finishes, physical distortion, corrosion of metals, entrapment of moisture, separation of bonded surfaces, or other physical damage.

3.7.4 Salt spray (corrosion) (see 4.6.5.4). The light source shall show no evidence of corrosive effects that impairs its specified functionality.

3.7.5 Fluid immersion (see 4.6.5.5). The light source shall reveal no evidence of swelling or softening of material, no loss of sealing capability, nor any other effects detrimental to the operation of the light source.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:33Z
Check the source to verify that this is the current version before use.

3.7.6 Shock (see 4.6.5.6). The light source shall show no evidence of physical damage that impairs its specified functionality.

3.7.7 Vibration (see 4.6.5.7). The light source shall show no evidence of physical damage that impairs its specified functionality.

3.7.8 Fungus resistance (see 4.6.5.8). The light source shall be in accordance with MIL-STD-454, requirement 4, for fungus-inert materials. Light sources that are not in accordance with requirement 4 of MIL-STD-454 shall meet grade I classification requirements of MIL-STD-810, method 508 when tested in accordance with 4.6.5.8.

3.7.9 Electromagnetic effects (see 4.6.5.9). The light source shall meet the conducted and radiated emission requirements and the conducted and radiated susceptibility requirements of MIL-STD-461, part 5, as specified (see 3.1).

3.8 Safety (see 4.6.1.1). The light source shall be in accordance with MIL-STD-454, requirement 1.

3.9 Transportability. The light source operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein.

3.10 Identification marking (see 4.6.1.1). The light source assembly shall be marked for identification in accordance with MIL-STD-130. Unless otherwise specified (see 3.1), the marking shall include the part number, the National Stock Number (NSN), Commercial and Government Entity (CAGE) number, classification code, manufacturer's name and date of manufacture.

3.11 Workmanship (see 4.6.1.1). All components shall be free from imperfections which will adversely affect the general appearance, function or serviceability. Surfaces of all parts shall be clean and free from extraneous materials. External surfaces shall be smooth and free from sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:33Z
Check the source to verify that this is the current version before use.

submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Inspection conditions shall be met as specified in the applicable test method for each inspection parameter, or as specified (see 3.1).

4.4 First article inspection. First article inspection shall consist of all the tests listed in table II. The units submitted for testing shall be representative of the production process.

TABLE II. First article inspection.

Inspection	Requirement	Test method	Sample size
Group I			
Visual and mechanical	3.5, 3.8, 3.10, 3.11	4.6.1.1	7
Weight	3.5.13	4.6.3.3	1/
Power supply voltage and current	3.6.1	4.6.4.1	1/
Light intensity	3.6.2	4.6.4.2	1/
Group II			
Terminal strength	3.5.11.1	4.6.3.1	1
Solderability	3.5.11.2	4.6.3.2	2/
Environmental temperature extremes	3.7.1	4.6.5.1	2/
Thermal shock	3.7.2	4.6.5.2	2/
Humidity	3.7.3	4.6.5.3	2/
Salt spray	3.7.4	4.6.5.4	2/

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection - Continued.

Inspection	Requirement	Test method	Sample size
Group II			
Fluid immersion	3.7.5	4.6.5.5	1
Shock	3.7.6	4.6.5.6	3/
Vibration	3.7.7	4.6.5.7	3/
Electromagnetic effects	3.7.9	4.6.5.9	3/
Fungus resistance	3.7.8	4.6.5.8	1
Maintainability	3.5.16	4.6.3.4	1
Reliability	3.3	4.6.2	3

1/ The same sample shall be used as in the visual and mechanical inspection.
2/ The same sample shall be used as in the terminal strength inspection.
3/ The same sample shall be used as in the fluid immersion inspection.

4.4.1 Sample. A light source sample shall be submitted for each light source construction for which first article approval is desired. The sample submitted shall be seven light sources of the same part number.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into five lots for group II testing. The sample units shall then be subjected to the inspection for their particular lot. After completion of sample testing, all units shall be resubjected to group I testing. Any light source failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be sufficient cause for refusal to grant first article approval. When first article test units are taken from a larger lot and the first article is disapproved, none of the units from that lot shall be delivered.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and tests specified for group A inspection (table III), group B inspection (table IV), and group C inspection (table V) (see 6.4).

4.5.1 Group A inspection. Group A inspections shall follow the order shown in table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:33Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection.

Inspection	Requirement	Test method
Visual and mechanical	3.5, 3.8, 3.10, 3.11	4.6.1.1
Weight	3.5.13	4.6.3.3
Power supply voltage and current	3.6.1	4.6.4.1
Light intensity	3.6.2	4.6.4.2

4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV. Group B inspections shall be made on units that have passed the group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Shock	3.7.6	4.6.5.6
Thermal shock	3.7.2	4.6.5.2
Terminal strength	3.5.11.1	4.6.3.1
Solderability	3.5.11.2	4.6.3.2

4.5.3 Group C inspections. Group C inspections shall consist of the inspections in table V. Group C inspections shall have satisfactorily completed all group A and group B inspections. After completion of group C inspections, the samples shall be re-subjected to group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

TABLE V. Group C inspection.

Inspection	Requirement	Test method
Environmental temperature extremes	3.7.1	4.6.5.1
Humidity	3.7.3	4.6.5.3
Salt spray	3.7.4	4.6.5.4
Fluid immersion	3.7.5	4.6.5.5
Shock	3.7.6	4.6.5.6
Vibration	3.7.7	4.6.5.7
Electromagnetic effects	3.7.9	4.6.5.9
Fungus resistance	3.7.8	4.6.5.8
Maintainability	3.5.16	4.6.3.4
Reliability	3.3	4.6.2

4.5.3.1 Disposition of inspection sample units. Sample units that have failed shall not be delivered.

4.6 Methods of inspection.

4.6.1 General testing practice.

4.6.1.1 Visual and mechanical examinations. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed using 5X magnification to verify that the design, construction, physical characteristics, dimensions, safety, identification marking, and workmanship are in accordance with the applicable requirements.

4.6.1.2 Standard test conditions. Unless otherwise specified (see 3.1), the following ambient conditions shall be used:

(a) Temperature: 25 plus 3, minus 5 degrees Celsius (°C).
(b) Relative humidity: 50 ± 30 percent.
(c) Absolute pressure: 725 ± 75 millimeters (mm) of mercury.

4.6.2 Reliability (see 3.3). Three light sources shall be tested in accordance with MIL-STD-781, appendix C, test plan 30-2. One light source shall be tested for 543 hours, another for 406 hours, and the third for 271 hours in order to achieve the required testing time. The light sources shall then be operated continuously for a minimum of 4 hours, using nominal voltage and at the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

maximum light intensity setting. The light sources shall then be tested in accordance with 4.6.4.2 to verify proper operation of the intensity control circuit.

4.6.3 Design and construction inspection.

4.6.3.1 Terminal strength (see 3.5.11.1). The terminals shall be tested in accordance with MIL-STD-202, method 211. Test condition A and test condition E shall be used. The load for test condition A shall be 2.25 kilograms (kg).

4.6.3.2 Solderability (see 3.5.11.2). The terminals shall be tested in accordance with MIL-STD-202, method 208.

4.6.3.3 Weight (see 3.5.13). The light source weight shall be tested using scales of sufficient range and accuracy appropriate for the specified weight and tolerance.

4.6.3.4 Maintainability (see 3.5.16). The light source shall be tested in accordance with MIL-STD-471, method 1.

4.6.4 Performance tests.

4.6.4.1 Power requirements. The maximum current shall be measured in accordance with MIL-STD-883, method 3005, using the nominal supply voltage specified. The power consumption shall then be calculated and verified to meet the requirements of 3.6.1.

4.6.4.2 Light intensity (see 3.6.2). The light intensity shall be verified at the light guide cable socket using a photometer at five check points, as marked on the intensity control: zero, one-quarter, one-half, three-quarters, and full intensity.

4.6.5 Environmental tests.

4.6.5.1 Environmental temperature extremes (see 3.7.1). The light source shall be tested in accordance with MIL-STD-810, method 501, procedure II, at the specified high operating temperature extreme. The light source shall be tested in accordance with MIL-STD-810, method 502, procedure II, at the specified low operating temperature extreme. The test duration shall be 96 hours. At the completion of the test, the light source shall be visually examined in accordance with 4.6.1.1.

4.6.5.2 Thermal shock (see 3.7.2). Light sources shall be tested in accordance with MIL-STD-810, method 503 for "Basic Hot (A2)", air deliverable, and "Cold (C2)" conditions. At the completion of the test, the light sources shall be visually examined in accordance with 4.6.1.1.

4.6.5.3 Humidity (see 3.7.3). Light sources shall be tested in accordance with MIL-STD-810, method 507, procedure II, nonhazardous/induced. The relative humidity shall be 90 percent, and the temperature range shall be the specified operating temperature range. At the completion of the test, the light source shall be visually examined in accordance with 4.6.1.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:33Z
Check the source to verify that this is the current version before use.

4.6.5.4 Salt spray (corrosion) (see 3.7.4). Light sources shall be tested in accordance with MIL-STD-810, method 509, for resistance to a 5 ± 1 percent concentration salt fog environment for a 48-hour period. At the completion of the test, the light source shall be visually examined in accordance with 4.6.1.1.

4.6.5.5 Fluid immersion (see 3.7.5). Light sources shall be tested for resistance to fluids specified in table VI. A portion of the light source outer cover, adhesive areas, and the light guide connection point shall be covered with each fluid listed in table VI. Each type fluid shall remain on each area for a period of 48 hours. At the completion of the test, the light source shall be visually examined in accordance with 4.6.1.1.

TABLE VI. Immersion test fluids.

Fluids	Specification	Test temperature (°C)
Fuel oil	MIL-F-16884	33 - 37
Turbine fuel (JP-4 and JP-5)	MIL-T-5624	20 - 25
Isopropyl alcohol	TT-I-735	20 - 25
Hydraulic fluids	MIL-H-5606 MIL-H-17672	48 - 50
Lubricating oils	MIL-L-17331 MIL-L-23699	73 - 77
Coolant 1/	-	20 - 25
Seawater	-	20 - 25
Tap water	-	20 - 25

1/ Monsanto Coolanol 25 or equivalent.

4.6.5.6 Shock (see 3.7.6). Light sources and accessories, in their carrying cases, shall be tested in accordance with MIL-STD-810, method 516, procedure IV. The light source and accessories, after exposure to the designated shock conditions, shall be removed from the carrying case and visually examined in accordance with 4.6.1.1.

4.6.5.7 Vibration (see 3.7.7). Light sources and accessories, in their carrying cases, shall be tested in accordance with MIL-STD-810, method 514, procedure I, categories 1 and 3. At the completion of the test, the light source shall be visually examined in accordance with 4.6.1.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:33Z
Check the source to verify that this is the current version before use.

4.6.5.8 Fungus resistance (see 3.7.8). Light sources and carrying cases that do not meet the requirements of fungus-inert materials in accordance with MIL-STD-454, requirement 4, shall be tested by exposure to fungus in accordance with MIL-STD-810, method 508.

4.6.5.9 Electromagnetic effects (see 3.7.9). The light source shall be tested for effects of electromagnetic emissions in accordance with MIL-STD-462 as applicable.

4.6.6 Inspection of packaging (see 5.1). The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5 and the marking requirements of MIL-STD-129.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 Preservation. Unless otherwise specified (see 6.2), the preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The light source shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Unit protection. The light source and accessories shall be placed in their respective storage compartments in the carrying case. All components, in the carrying case, shall be securely cushioned with materials conforming to PPP-C-843, type II. Each component shall be packaged in accordance with MIL-P-116, method IC-2. Carrying case fasteners and hinges shall be taped over or cushioned before placing in the unit box in accordance with MIL-P-116.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The light source is intended for use with rigid and flexible fiberscopes. It will provide high-intensity light to the fiberscope for visual inspection of internal equipment areas.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Applicable specification sheet number, title, and date.
(d) When first article is required (see 3.2).
(e) Applicable part number (see 6.7).
(f) Quantity of light sources required.
(g) Selection of applicable levels of preservation and packing (see 5.1).

6.3 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of 8 units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.5 Definitions.

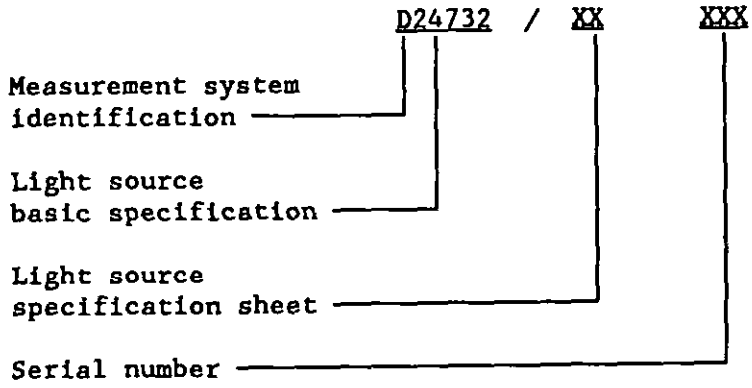
6.5.1 Fiberscope light source. A fiberscope light source is a device used to provide effective illumination over the fiberscope's field of view.

6.5.2 Light intensity. Light intensity is the radiant power emitted per unit area of a source.

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:33Z
Check the source to verify that this is the current version before use.

6.7 Part or Identifying Number (PIN). Light sources should contain only the following:



Examples:

D24732/01-001
D24732/01-002

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Subject term (key word) listing.

Borescope light source
Endoscope light source
Fiberscope
Inspection scope light source
Light intensity

Review activities:
Navy - AS, EC, YD

Preparing activity:
Navy - SH
(Project 6032-N001)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-24732(NAVY)	LIGHT SOURCE, RIGID AND FLEXIBLE FIBERSCOPE, FIBER OPTIC (METRIC), GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:33Z
Check the source to verify that this is the current version before use.